Problem Setting: Time-Budgeted Value-Maximizing Transit Itinerary Planning

The task is not to find the earliest-arrival journey, but to construct a timetable-feasible route that collects more itinerary value within a fixed budget.

1. Query and Timetable Constraints

Input: origin stop, departure time, and time budget.

Every route must satisfy GTFS departures, arrivals, transfer buffers, and budget limits.

2. Value to Be Maximized

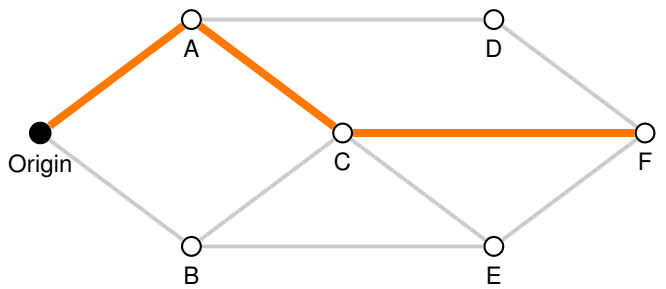
Instead of minimizing travel time only, we maximize a multi-metric route value: ride duration, unique stops, geographic spread, and quadrant coverage.

3. Desired Output

A feasible itinerary that explores the city more broadly within the same budget, rather than simply ending as early as possible.

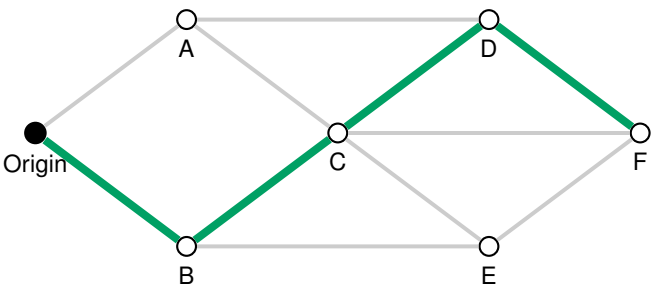
Shortest-path query vs. **value-maximizing itinerary**

Conventional earliest-arrival style



Ride = short | Stops = few | Spread = narrow

Ours: value-maximizing itinerary



Ride = longer | Stops = more | Spread = broader

The proposed method keeps timetable feasibility unchanged but changes the optimization target from shortest journey to budgeted itinerary value.